

UNIT-3 SOLVED PROBLEMS

COURSE NAME:FUNDAMENTALS OF DATA SCIENCE

COURSE CODE:DSA0404

NAME:SANIA HERBERT

REG NO:192324062

1. Mean, Median, Mode and Range

Data:

12, 15, 18, 15, 20, 22, 15, 25

Step 1: Arrange the data

12, 15, 15, 15, 18, 20, 22, 25

Mean

Formula:

Mean: $\sum X/N$

Number of observations: N=8

Therefore:

Mean= $142/8=17.75$

Median

There are 8 observations, so the median is the average of the 4th and 5th values.

Median= $15+18/2$

median=16.5

Mode

The value occurring most frequently is 15.

15 occurs 3 times.

Mode=15

Range

=Maximum-Minimum

=25-12=13

Que 2. Frequency Distribution

Data:

2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4

We count how many times each value occurs.

Value	Frequency
2	3
3	3
4	3
5	2
6	1

Value	Frequency
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Total 12

Verification

$$3+3+3+2+1=12$$

So the frequency distribution is correct.

Relative Frequency

Relative frequency: frequency/total observations

Value	Frequency	Relative Frequency
2	3	25%
3	3	25%
4	3	25%
5	2	16.67%
6	1	8.33%

Total 12 100%

Therefore, 2, 3 and 4 are the most frequent values.

Que 3. Q1, Q2, Q3 and IQR

Data:

5, 8, 10, 12, 15, 18, 20, 22, 25, 30

The data is already sorted.

There are: $n=10$

Q2 — Median

Since there are 10 values:

$$Q2 = 5^{\text{th}} \text{ value} + 6^{\text{th}} \text{ value} / 2$$

$$Q2 = 15 + 18 / 2$$

$$Q2 = 16.5$$

Q1

Lower half:

5, 8, 10, 12, 15

The median of these five values is: **Q1=10**

Q3

Upper half:

18, 20, 22, 25, 30

The median of these five values is: **Q2=22**

IQR

Formula: IQR=Q3-Q1

$$IQR = 22 - 10$$

$$IQR = 12$$

Que: 4. Detect Outliers Using IQR Method

Data:

10, 12, 13, 15, 16, 17, 18, 20, 45

The data is already sorted.

There are 9 observations.

Step 1: Find Q1

Lower half, excluding the median:

10, 12, 13, 15

$$\mathbf{Q1=12+13/2}$$

$$\mathbf{Q1=12.5}$$

Step 2: Find Q3

Upper half:

17, 18, 20, 45

$$\mathbf{Q3=18+20/2}$$

$$\mathbf{Q3=19}$$

Step 3: Calculate IQR

Step 4: Calculate Lower Fence

Formula:

$$\text{Lower Fence} = Q_1 - 1.5(IQR)$$

$$= 12.5 - 1.5(6.5)$$

$$= 12.5 - 9.75$$

$$\boxed{\text{Lower Fence} = 2.75}$$

Step 5: Calculate Upper Fence

$$\text{Upper Fence} = Q_3 + 1.5(IQR)$$

$$= 19 + 1.5(6.5)$$

$$= 19 + 9.75$$

$$\boxed{\text{Upper Fence} = 28.75}$$

Any value:

- below 2.75 $\rightarrow$ outlier
- above 28.75 $\rightarrow$ outlier

The value 45 is greater than 28.75.

Therefore:

45 is an outlier

5. Determine Skewness

Data:

5, 6, 7, 7, 8, 8, 9, 20

There is an unusually high value, 20, compared with the remaining observations.

Step 1: Calculate Mean

$$\begin{aligned}\text{Mean} &= \frac{5 + 6 + 7 + 7 + 8 + 8 + 9 + 20}{8} \\ &= \frac{70}{8}\end{aligned}$$

$$\text{Mean} = 8.75$$

Step 2: Calculate Median

There are 8 observations.

The middle values are 7 and 8.

$$\text{Median} = \frac{7 + 8}{2}$$

$$\text{Median} = 7.5$$

Step 3: Mode

7 occurs twice and 8 occurs twice.

Therefore the dataset is bimodal:

$$\text{Modes} = 7, 8$$

Determine the skewness

We observe:

$$\text{Mean} > \text{Median}$$

$$8.75 > 7.5$$

This indicates positive skewness (right skewness).

The value 20 creates a long tail on the right side.

Final

Answer

Distribution is positively/right skewed

Que: 6. Uniform Distribution

A uniformly distributed variable ranges from:

$$a = 10, \quad b = 50$$

Find:

For a continuous uniform distribution:

$$P(c < X < d) = \frac{d - c}{b - a}$$

Therefore:

$$\begin{aligned} P(20 < X < 35) &= \frac{35 - 20}{50 - 10} \\ &= \frac{15}{40} \\ &= 0.375 \end{aligned}$$

As a percentage:

$$0.375 \times 100 = 37.5\%$$

Answer

$$P(20 < X < 35) = 0.375$$

Que:7. Find Z-score

Given:

$$\mu = 100$$

Mean:

$$\sigma = 15$$

Standard deviation:

$$X = 130$$

Value:

$$Z = \frac{X - \mu}{\sigma}$$

Formula:

Substitute:

$$Z = \frac{130 - 100}{15}$$

$$Z = \frac{30}{15}$$

$$\boxed{Z = 2}$$

Interpretation

The value 130 is 2 standard deviations above the mean.

Que:8. Find $P(X < 130)$

We already calculated : $Z=2$

Therefore : $P(X < 130) = P(Z < 2)$

From the standard normal distribution table:

$$P(Z < 2) = 0.9772$$

Therefore:

$$P(X < 130) = 0.9772$$

As a percentage: $P(X < 130) = 97.72\%$

Answer

This means approximately 97.72% of observations are expected to be below 130.

Que:9. Compare Mean and Median Before and After Outlier Treatment

Consider the dataset:

10, 12, 13, 14, 15, 16, 18, 100

Here, 100 is an extreme outlier.

Before Outlier Treatment

Mean

$$\begin{aligned}\text{Mean} &= \frac{10 + 12 + 13 + 14 + 15 + 16 + 18 + 100}{8} \\ &= \frac{198}{8}\end{aligned}$$

$$\boxed{\text{Mean} = 24.75}$$

Median

There are 8 values.

$$\text{Median} = \frac{14 + 15}{2}$$

$$\boxed{\text{Median} = 14.5}$$

After Outlier Treatment

Suppose we remove the outlier 100.

Remaining data:

10, 12, 13, 14, 15, 16, 18

Mean

$$\begin{aligned}\text{Mean} &= \frac{10 + 12 + 13 + 14 + 15 + 16 + 18}{7} \\ &= \frac{98}{7} \\ \boxed{\text{Mean} = 14}\end{aligned}$$

Median

There are 7 observations.

The middle value is: Median=14

Comparison

Measure	Before	After
Mean	24.75	14
Median	14.5	14

Conclusion

The outlier 100 strongly affected the mean, increasing it from 14 to 24.75.

The median changed only slightly:

This demonstrates an important property of statistics:

Mean is sensitive to extreme values, while median is relatively resistant to outliers.

Que:10. Complete EDA on a Small Student-Mark Dataset

Let's use the following student marks dataset:

Student	Marks
S1	45
S2	50
S3	55
S4	60
S5	62
S6	65
S7	68
S8	70
S9	72

Student	Marks
S10	95

We will analyze:

1. Central tendency
2. Dispersion
3. Distribution
4. Outliers
5. Skewness
6. Overall conclusion

A. Basic Information

Number of students: $n=10$

Minimum:45

Maximum:95

B. Central Tendency

Mean

$$\text{Mean} = \frac{45 + 50 + 55 + 60 + 62 + 65 + 68 + 70 + 72 + 95}{10}$$

Sum:

$$642$$

Therefore:

$$\text{Mean} = \frac{642}{10}$$

$$\boxed{\text{Mean} = 64.2}$$

Median

There are 10 observations.

The middle values are the 5th and 6th values:

$$\text{Median} = \frac{62 + 65}{2}$$

$$\boxed{\text{Median} = 63.5}$$

Therefore:

Mode

Every value occurs only once.

Therefore: there is no mode

Central Tendency Summary

Measure	Result
Mean	64.2
Median	63.5
Mode	No mode

Since:

there is an indication of slight positive skewness.

C. Dispersion

Range

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

$$= 95 - 45$$

$$\boxed{\text{Range} = 50}$$

Q1 and Q3

Lower half:

45, 50, 55, 60, 62

Therefore: Q1=55

Upper half:

65, 68, 70, 72, 95

Therefore: $Q_3=70$

IQR

$$IQR = Q_3 - Q_1$$

$$= 70 - 55$$

$$\boxed{IQR = 15}$$

Variance and Standard Deviation

Using the population formula:

$$\sigma^2 = \frac{\sum (X - \mu)^2}{N}$$

For this dataset:

$$\boxed{\text{Variance} = 183.96}$$

Therefore:

$$\sigma = \sqrt{183.96}$$

Standard Deviation ≈ 13.56

Dispersion Summary

Measure	Value
Minimum	45
Maximum	95
Range	50
Q1	55
Q3	70
IQR	15
Variance	183.96
Standard Deviation	≈ 13.56

D. Outlier Detection

Use the IQR method.

$$Q_1 = 55$$

$$Q_3 = 70$$

$$IQR = 15$$

Lower Fence

$$Q_1 - 1.5(IQR)$$

$$= 55 - 1.5(15)$$

$$= 55 - 22.5$$

$$\boxed{32.5}$$

Upper Fence

$$Q_3 + 1.5(IQR)$$

$$= 70 + 1.5(15)$$

$$= 70 + 22.5$$

$$\boxed{92.5}$$

Any observation above 92.5 is an outlier.

E. Distribution Analysis

The marks are:

45, 50, 55, 60, 62, 65, 68, 70, 72, 95

Most values are concentrated between approximately 45 and 72, while 95 is considerably higher.

Therefore, the distribution has a longer right tail.

F. Skewness Interpretation

A common interpretation is:

- Mean $\approx$ Median $\rightarrow$ approximately symmetric
- Mean $>$ Median $\rightarrow$ positively/right skewed
- Mean $<$ Median $\rightarrow$ negatively/left skewed
- The high mark of 95 contributes to this skewness.

G. Overall EDA Report

Category	Finding
Number of students	10
Minimum mark	45
Maximum mark	95
Mean	64.2
Median	63.5
Mode	No mode
Range	50
Q1	55
Q3	70
IQR	15
Variance	183.96
Standard deviation	≈ 13.56
Lower fence	32.5
Upper fence	92.5
Outlier	95

Category	Finding
Distribution	Right-skewed
Skewness	Positive

Final Interpretation

The student dataset has an average mark of 64.2, with a median of 63.5. The relatively close mean and median indicate that the dataset is not extremely skewed. However, the mark 95 is identified as an outlier using the IQR method because it exceeds the upper fence of 92.5.

The standard deviation of approximately 13.56 indicates a moderate amount of variation in student performance. The range of 50 marks also shows a substantial difference between the lowest and highest scores.

Overall, the distribution is slightly positively skewed, mainly because of the high score of 95.